



UITs

**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

Assignment

Course Title: Simulation and Modeling Lab
Course Code: CSE 413

Submitted by:	Submitted to:
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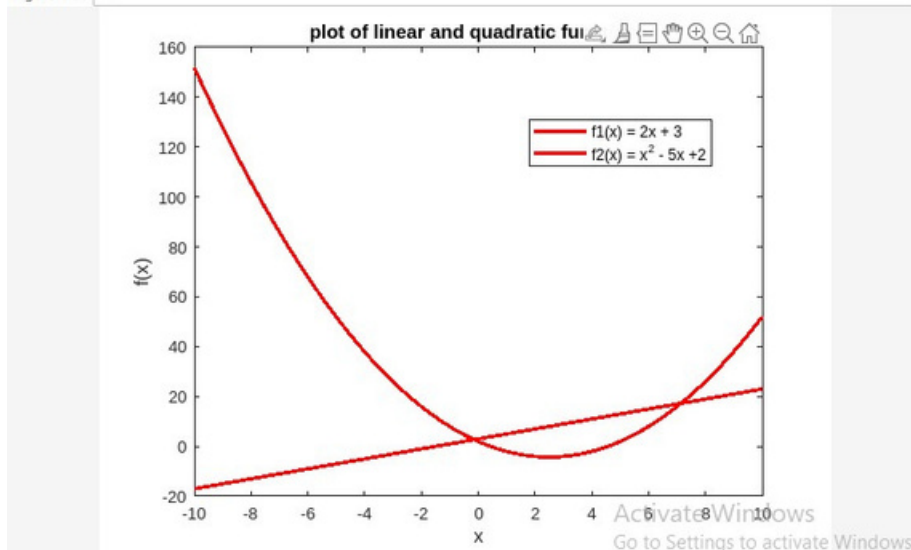
1. Write a MATLAB script to define and plot the following

functions for x values from -10 to 10:

- Linear function: $f_1(x) = 2x + 3$
- Quadratic function: $f_2(x) = x^2 - 5x + 2$

Use different colours and line styles for each function. Add axis labels, a title, and legend to the plot. Identify the point where both functions intersect.

```
Assi_q1.m x Assi_q2.m x +
/MATLAB Drive/Assi_q1.m
1 x = -10:0.1:10;
2
3 f1 = 2*x + 3;
4 f2 = x.^2 - 5*x + 2;
5 %the plot
6 figure;
7 plot(x, f1, 'Color', 'r', 'LineStyle', '-', 'LineWidth', 2);
8 hold on;
9 plot(x, f2, 'Color', 'r', 'LineStyle', '-', 'LineWidth', 2);
10
11 title('plot of linear and quadratic functions');
12 xlabel('x');
13 ylabel('f(x)');
14
15 legend('f1(x) = 2x + 3', 'f2(x) = x^2 - 5x + 2', 'Location', 'best');
16
17 plot(x_intersect, y_intersect, 'Marker', 'o', 'MarkerSize', 8, 'MarkerEdgeColor', 'k', 'MarkerFaceColor', 'k');
18 fprintf('The point where both functions intersect is (%.2f, %.2f)\n', x_intersect, y_intersect);
19
20
21
```



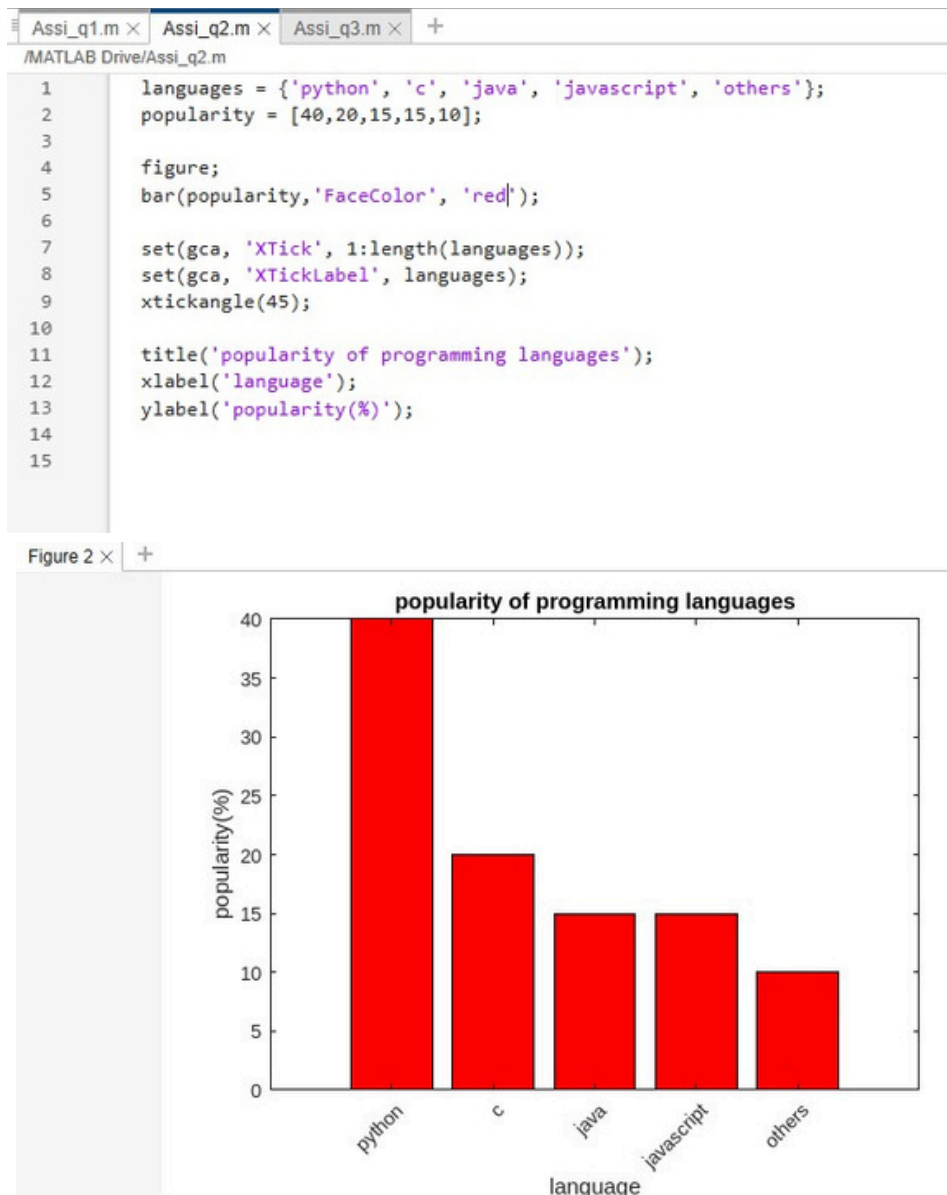
Function Plotting:

This code plots a linear function ($2x + 3$) and a quadratic function ($x^2 - 5x + 2$) on the same graph, then calculates and displays the intersection points of those two functions.

2. Consider a dataset representing the popularity of different programming languages based on usage:

- Python: 40%, C: 20%, Java: 15%, JavaScript: 15%, Others: 10%

Write a MATLAB script to create a bar chart to visualise the data. Label each bar properly and use different colours. Add a title, axis labels, and a legend.



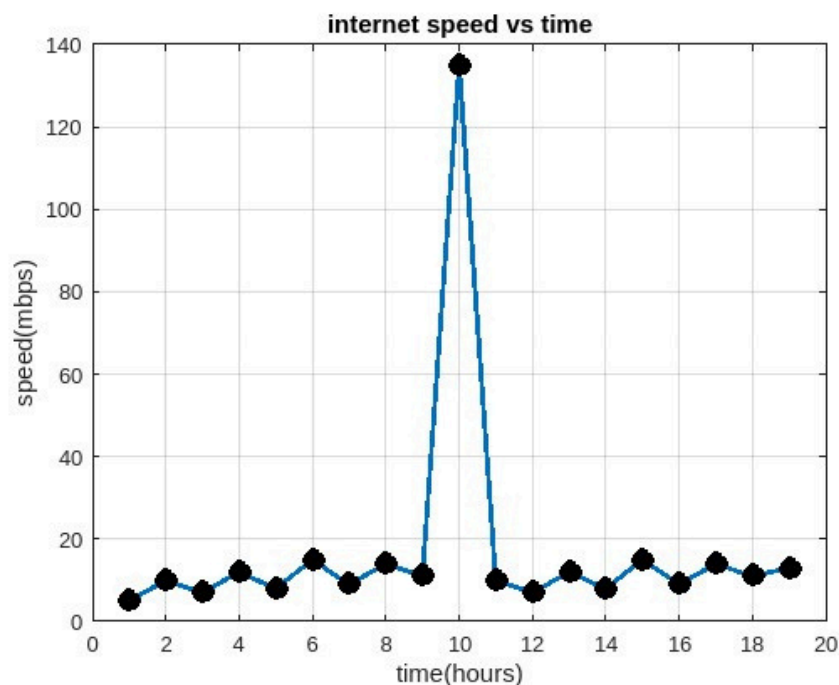
Bar Chart of Language Popularity:

This code creates a bar chart visualizing the popularity of different programming languages. The height of each bar represents the popularity percentage of a language. It also labels and titles the chart for clarity.

3. Assume internet speed is measured every hour for 10 hours,

and the recorded speeds (in Mbps) are:5,10,7,12,8,15,9,14,11,135,10,7,12,8,15,9,14,11,13

Write a MATLAB script to plot the internet speed vs. time using a line plot. Use markers to highlight each data point. Add labels for time (x-axis) and speed (y-axis), along with a title. Analyze whether the speed is stable or fluctuating.



Internet Speed vs. Time Plot:

This code generates a line plot showing internet speed (in Mbps) over a 19-hour period. It marks each data point and calculates the mean and standard deviation of the speed data, highlighting a clear outlier.